

Standard Operating Procedure
 End-to-End Operation of the agentic-bioinspired-cad
 Generative Design & Single-Material Print Loop
 Document ID: ABCAD-SOP-001 Revision B

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Document control

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Revision history

Rev.	Date	Description of change
A	2026-07-03	Initial release.
B	2026-09-25	Public release. Identifiers renamed to the <code>abcad</code> namespace; host-specific paths replaced by <code>ABCAD_*</code> environment variables; procedure updated to the current loop (in-loop STL export, deep print gate with deterministic auto-scale, schema-constrained critic verdicts, run manifest); enamel decussation family added to the scope.

1 Purpose

This SOP defines the controlled, repeatable procedure for operating the agentic-bioinspired-cad pipeline from a natural-language design request to a *certified-printable* single-material part: local LLM code generation → Blender geometry realization and rendering → vision-language-model (VLM) critique → automated code repair and design refinement → manufacturability gating with a quantitative print audit → printable variant preparation → (optional) finite-element verification → slicing and printing. It specifies pre-conditions, per-stage acceptance criteria, system-protection limits, nonconformance handling, and the records that constitute run evidence.

2 Scope

2.1 In scope

Operation of the loop on a workstation meeting Section 6 for the three structure families of the package (helicoidal/Bouligand laminates, woven lattices, enamel decussation lattices); print preparation for fused-deposition modeling (FDM, 0.4 mm nozzle) and stereolithography (resin) processes; the associated verification tooling (Blender-free test suites, print audit, linear and finite-strain FEA cross-checks).

2.2 Out of scope

Model training and LoRA fine-tuning; printer hardware maintenance and calibration; resin handling and post-cure chemistry; fracture simulation (covered by test method TM-6 of ABCAD-QMS-002); multi-material printing (prohibited by design constraint C1, Section 4).

3 References

- ABCAD-QMS-002 — *Quality Definitions & Test Methods* (vocabulary, test methods TM-1 to TM-6, validity conditions).
- Repository documentation: `docs/WORKFLOWS.md` (per-structure runbook), `docs/PRINTING.md` (print audit), `docs/FEA.md` (FEA instruments), `docs/PERF.md` (resource envelope), and the agent documentation in `abccad/agent/`.
- Evidence index: `results/README.md` (curated records behind the reference values of Appendix C).

4 Definitions and design constraints

LoRA Low-rank adaptation of Llama-3.2-3B-Instruct emitting Blender-Python “code-as-geometry” (the Phase-1 code emitter).

VLM Vision-language model critic (qwen3-v1:8b via Ollama) grading renders against retrieved reference images and returning a schema-constrained JSON verdict.

RAG Retrieval-augmented grounding. Text corpus of code exemplars (`abccad/data/rag/text.corpus.jsonl`); image corpus of reference renders (`abccad/data/rag/vlm.corpus.jsonl` with images in `abccad/data/rag/renders/`).

EDT Euclidean distance transform; basis of the wall-thickness / clearance audit (morphological opening at diameter d : erosion then dilation by $d/2$).

E_{eff} Effective uniaxial modulus of a lattice specimen, $E_{\text{eff}} = (F/A_0)/\varepsilon$, with reaction force F , nominal footprint A_0 , nominal strain ε . Scale-invariant for geometrically similar specimens.

Watertight / manifold Zero boundary edges / zero non-manifold edges on the triangulated surface.

C1 (constraint) Single material only. All mechanical function derives from geometry/topology; never a hard/soft material contrast. Literature toughness numbers obtained with a large modulus contrast (e.g. Jia & Wang 2019, $\sim 390\times$) must not be transferred.

C2 (constraint) Memory safety first. On a 16 GB host the limits of Section 7 are mandatory, not advisory.

5 Responsibilities

Role	Duties under this SOP
Operator	Authorizes heavy runs; establishes the memory baseline (§7); executes the pipeline stages; enforces watchdog limits; owns printer operation, slicing, and physical QC; records results; approves deviations; halts and dispositions any acceptance-criteria failure.
Automated subsystems	Code emitter (LoRA), code fixer and code designer (<code>qwen2.5-coder:7b</code>), VLM critic (<code>qwen3-vl:8b</code>), manufacturability gate and print audit — operate only within the loop’s guardrails.

6 System, equipment, and materials

6.1 Host

Reference host: Apple-silicon macOS workstation with **16 GB unified memory** (the limiting resource). All commands are issued from the repository root; interpreters and tools are resolved from the environment variables below, and every tool writes under `$ABCAD_OUT`.

6.2 Software environments

Component	Specification
<code>ABCAD_CAD_PYTHON</code>	Interpreter of the conda CAD environment (<code>env/cad_env.lock.yml</code>): numpy, scipy, pyvista/vtk, numpy-stl, cadquery, and the loop’s model and retrieval stack. Runs builds, audits, renders, retrieval, and the Phase-2 parent process.
<code>ABCAD_SFEPY_PYTHON</code>	Interpreter of the conda SfePy environment (<code>env/sfepy_env.lock.yml</code>): sfepy, numpy 2, pyang. Runs <i>all</i> FEA. Invoke with <code>-u</code> (unbuffered progress).
<code>ABCAD_BLENDER</code>	Blender 5.x executable (EVEE renderer), run headless.
Ollama	Local model server with <code>qwen2.5-coder:7b</code> (4.7 GB) and <code>qwen3-vl:8b</code> (6.1 GB). <i>Not</i> running by default; launched capped per §8.2.
Code emitter	Llama-3.2-3B-Instruct + LoRA on the Apple GPU (float16), ~8–9 GB class, subprocess-isolated (Phase 1 exits before Phase 2 loads any model).
<code>ABCAD_OUT</code>	Output root for every tool (default <code>./out</code>); never the package directory.

6.3 Print processes and limits

One set of process limits is shared by the in-loop manufacturability gate and the print audit (`abcad/printing/print_audit.py`); the process is selected with `ABCAD_PRINT_PROCESS` (default `fdm`).

Process	Min. feature / strand	Overhang gate	Build volume [mm]
FDM, 0.4 mm nozzle (default)	≥ 0.8 mm	$\leq 15\%$ area $> 45^\circ$	$220 \times 220 \times 250$
Resin SLA/DLP	≥ 0.3 mm	n/a (supports)	$145 \times 145 \times 175$

Audit ladders: thickness diameters {0.3, 0.5, 0.8, 1.2} mm; clearance gaps {0.3, 0.5, 0.8} mm; a part “forms at diameter d ” if the morphological opening at d removes $\leq 1\%$ of its material.

7 Safety and system protection (mandatory)

The host is memory-constrained; an out-of-memory stall is the primary hazard (constraint C2).

1. **Pre-flight memory baseline before ANY heavy step** (code emitter, Ollama-backed loop, FEA above a few thousand hexahedra, MOOSE): close large applications and release inactive memory (on macOS, `sudo purge`); confirm that swap usage is near its baseline.
2. **Watchdog on every heavy step** (Appendix B). It samples swap usage and child-process memory every 2 s and terminates the process group on a threshold breach or time cap. Reference thresholds: 3072–3584 MB of swap for renders, FEA, and audits; 4096–5120 MB for the code emitter and the Ollama loop; time caps 6–45 min by step.
3. **Ollama caps**: `OLLAMA_MAX_LOADED_MODELS=1`, `OLLAMA_NUM_PARALLEL=1`, short keep-alive (2 min). Never allow two large models to be resident at once; respect the two-phase design (the generation child exits before critic models load).
4. **FEA mesh cap**: with the direct solver, $\leq 15\text{--}20\text{ k}$ hexahedra for the nonlinear solver (a 35 k-hex run exceeded 39 min and was terminated). Reference: 17,685 hex $\approx 1.65\text{ GB}$ resident, $\sim 1\text{ min}$. Larger linear meshes use `ABCAD_FEA_SOLVER=amg` (63,334 hex in 60 s).
5. **Abort posture**: on any watchdog trip, treat the run as failed (§10); never retry without re-establishing the memory baseline.

8 Procedure

Stages 8.1–8.9 constitute one controlled run. Do not skip gates; record outcomes per §8.9. In the listings, `<watchdog>` stands for the memory watchdog of Appendix B and `<agent>` for the command-line entry point of the design loop documented in `abccad/agent/`.

8.1 Stage 1 — Pre-flight verification (memory-light)

```
# baseline memory picture (record swap 'used' in the run log)
sysctl -n vm.swapusage ; vm_stat | head -6

# logic verification WITHOUT Blender or models
python -m pytest -q -m "not slow"
```

Acceptance criteria: All fast tests pass. Swap usage at or near its post-purge baseline ($\lesssim 1.5\text{ GB}$) before proceeding to any heavy stage; otherwise establish the baseline first (§7).

8.2 Stage 2 — Backend startup

```
OLLAMA_MAX_LOADED_MODELS=1 OLLAMA_NUM_PARALLEL=1 OLLAMA_KEEP_ALIVE=2m \
OLLAMA_CONTEXT_LENGTH=8192 \
  ollama serve & # capped local model server, 8k context
ollama list # verify the models are present
```

Acceptance criteria: `ollama list` shows `qwen2.5-coder:7b` and `qwen3-v1:8b`. The daemon idles (no model loaded) until the loop calls it. `OLLAMA_CONTEXT_LENGTH=8192` is **mandatory**: the critic prompt is $\sim 3.5\text{ k}$ tokens and the critic model's chat template enables thinking; at the default 4096-token context the verdict is truncated or lost.

8.3 Stage 3 — Generation and the render/critique/repair loop

Full end-to-end run (Phase-1 code emitter child process → Phase-2 loop), under the watchdog:

```
<watchdog> 5120 2700 run_swap.log -- <agent> "a woven cubic lattice metamaterial"
```

Reproducible loop testing and regression (injects saved code and skips the stochastic generator):

```
<watchdog> 5120 2700 run_swap.log -- <agent> --use-code saved_code.py "a woven cubic lattice metamaterial"
```

Expected controlled behavior:

- A failed initial render routes directly to the code fixer (no critic call on a missing image).
- Code fixer: ≤ 3 attempts; each log line shows the error under repair; verbatim repeats are detected *before* rendering and escalate the sampling temperature $0.1 \rightarrow 0.5 \rightarrow 0.8$ with an explicit restructure demand; a repeat at maximum temperature fails fast. Reference performance: a structural mesh/curve API error repaired in 2 attempts (reproduced $3\times$).
- Critic: the verdict JSON is printed (`match_quality`, `physical_stability`, comment); a failed critique retries once and is marked **VLM CRITIQUE FAILED** — it must not be treated as a design critique, and the run ends rather than designing on it.
- Code designer: parameter-level changes only; a designer edit that breaks execution routes to the code fixer (never crashes the graph); verbatim-repeat designs trip the stall guard and end the run.
- Manufacturability gate: process from `ABCAD_PRINT_PROCESS` (default `fdm`), gating on the world bounding box and the $45^\circ/15\%$ overhang rule from the render-time mesh statistics. After approval, the full print audit runs once on the exported STL (deep gate); a thin-feature-only failure is repaired deterministically by uniform scaling with the computed factor k , and the scaled variant is re-audited.
- Every successful render exports the scene geometry to STL; the terminal record is `run_manifest.json` (prompt, verdicts, artifact paths including scaled variants, counters).

Acceptance criteria: Run terminates without crash; `blender_status=success`; critic `approve=true` with `match_quality=good` and `physical_stability=stable`; manufacturability `printable=true`, or a PRINT-certified auto-scaled variant recorded in the manifest. Any other terminal state → §10.

8.4 Stage 4 — Geometry export

The loop exports the geometry of every successful render; the manifest records the approved STL (`stl_path`) and any scaled variant (`stl_path_scaled`). Deterministic generator families produce STL natively (`docs/GENERATORS.md`). Fallback for a saved script:

```
"$ABCAD_BLENDER" -b --python-expr "import bpy
exec(open('generated.py').read())
try: bpy.ops.wm.stl_export(filepath='part.stl')
except Exception: bpy.ops.export_mesh.stl(filepath='part.stl')"
```

Acceptance criteria: STL written; the next stage's audit (not this step) is the arbiter of its quality.

8.5 Stage 5 — Quantitative print audit

```
<watchdog> 3584 1800 audit_swap.log -- \
"$ABCAD_CAD_PYTHON" -u -m abcad.printing.print_audit \
--stl part.stl # add --voxel 0.15 to certify a chosen print candidate
```

The audit measures: watertight/manifold; bounding box, solid volume, resin-equivalent mass; bed-aware 45° overhang fraction; EDT wall-thickness ladder (maximum certified diameter; lost-material fraction per rung); near-structure clearance / fiber-fusion ladder; enclosed voids (trapped resin); per-process verdicts with concrete fixes (supports, drain holes, scale factor). Reports land in \$ABCAD_OUT/print_audit/.

Acceptance criteria: For the default FDM process: watertight = manifold = true; lost-material fraction at $0.8\text{ mm} \leq 1\%$; overhang $\leq 15\%$ *or* an explicit supports plan; bounding box within $220 \times 220 \times 250\text{ mm}$. Trust floor: thickness statements are valid down to ~ 2 voxels at the reported voxel size — re-run with `--voxel 0.15` before committing a print.

8.6 Stage 6 — FDM variant preparation (when Stage 5 fails on thin features)

Uniform scaling preserves the architecture exactly; E_{eff}/E is scale-invariant, so prior FEA on the original remains valid for the scaled part. Scale factor rule: $k \geq d_{\text{limit}}/d_{\text{certified}}$ (from the Stage-5 report), rounded up to 0.1.

```
"$ABCAD_CAD_PYTHON" -m abcad.printing.fdm_variants \
--stl part.stl --scales 1.5,2.0 --plate 2.0 --overlap 0.8
```

`--plate` additionally emits a plated tension specimen (grip plates as overlapping closed shells, FEA plating convention: plate thickness and bonded overlap given at $\times 1$ scale and scaled with the part; slicers union the shells). Open-tube lattices that fail the watertight check are remeshed as solids with `abcad.printing.remesh_watertight`.

Acceptance criteria: Re-run Stage 5 on every variant. Reference certification (29 mm woven specimen): $\times 1.5$ = FDM PRINT (marginal, $0.99\% < 0.8\text{ mm}$); $\times 2.0$ = FDM PRINT (clean, 0.50%); $\times 2.0$ plated = FDM PRINT with fixes (top-plate underside, 16.7% overhang).

8.7 Stage 7 — Mechanical verification by FEA (optional, per design intent)

```
# mesh: STL -> plated voxel-hex (+ A0/H0 sidecar); keep <= ~15-20k hex for the direct
  solver
"$ABCAD_CAD_PYTHON" -m abcad.fea.voxel_plate_mesh \
--stl part.stl --voxel 1.0 --plate 2.0 --overlap 0.8 --msh part.msh --no_render

# linear reference (trusted): E_eff (ABCAD_FEA_SOLVER=amg for large meshes)
"$ABCAD_SFEPY_PYTHON" -u abcad/fea/run_tension.py --msh part.msh

# finite strain (neo-Hookean total-Lagrangian, strain control)
"$ABCAD_SFEPY_PYTHON" -u -m abcad.fea.run_tension_finite --msh part.msh

# solver self-check whenever run_tension_finite is modified
"$ABCAD_SFEPY_PYTHON" -u -m abcad.fea.run_tension_finite --block 10,10,20,5,5,9
```

Acceptance criteria: Block validation recovers the solid modulus within $\sim 5\%$ (reference: $3080.8 \text{ MPa} = 102.7\%$ of $E = 3000 \text{ MPa}$, concave neo-Hookean response). Lattice finite-strain E_{eff} agrees with the linear reference on the same mesh to within a few percent at small strain (reference, 17,685-hex woven mesh: 93.1 vs 91.8 MPa , $+1.4\%$; interior residual $\sim 10^{-7} \text{ N}$). Never trust an FEA conclusion that fails to recover the linear reference.

8.8 Stage 8 — Slicing and printing

- Orientation: as modeled ($+z$ up) unless the Stage-5 report says otherwise; the audit's overhang number applies to that orientation only.
- Reference defaults (FDM, 0.4 mm nozzle): 0.2 mm layers; brim on (lattice feet are small stubs); 100% infill for specimens compared with FEA; supports per the audit verdict (bare woven lattice at $\times 2.0$: none; plated specimen: bridging or supports under the top plate); part cooling high at fiber crossings.
- Slice, review the preview layer by layer at the first fiber-crossing height, print.

Acceptance criteria: Physical QC of the printed part: mass within $\pm 15\%$ of the audit's volume $\times$ material density (sliced solid fibers); overall bounding box within $\pm 0.5 \text{ mm}$ of the STL; no fused-solid regions where the clearance ladder predicted open gaps; fibers continuous (no missing strands).

8.9 Stage 9 — Records and closeout

Record in the run log: prompt, run folder, per-stage outcomes with their numbers, memory peaks from the watchdog logs, deviations and dispositions. Artifacts retained: run folder (renders, generated code, `run_manifest.json`), print-audit reports, exported and variant STLs, FEA JSON/CSV, sliced project file. Records that back published numbers are archived under `results/`.

Acceptance criteria: The run log allows a third party to reconstruct the run and its evidence without any other source.

9 Quality-control summary (all gates at a glance)

Gate	Criterion	Method	Record
Logic	fast test suite passes	Blender-free tests	run log
Memory baseline	swap used $\lesssim 1.5 \text{ GB}$ before heavy steps	<code>sysctl vm.swapusage</code>	run log
Runtime safety	no watchdog trip (swap $<$ threshold; within time cap)	watchdog log	<code>*_swap.log</code>
Render	validation success; mesh statistics parsed	headless Blender validation	run folder
Critique	critic <code>approve=true</code> (good + stable)	<code>qwen3-v1</code> vs references	manifest
Gate (in loop)	bbox within bed; overhang $\leq 15\%$ (FDM)	manufacturability gate	manifest
Geometry	watertight; manifold	print audit surface checks	audit CSV/MD

Thickness	lost fraction at process limit $\leq 1\%$	EDT opening	morphological	audit CSV/MD
Clearance	fusion-risk fractions reviewed; enclosed voids positioned	EDT of the void phase		audit CSV/MD
FEA (optional)	finite $\approx$ linear at small strain; block $\approx E$	§8.7		FEA JSON/CSV
Physical part	mass $\pm 15\%$; ± 0.5 mm; gaps open; strands continuous	bbox bench check		run log

10 Nonconformance, deviations, and troubleshooting

On any acceptance-criteria failure: STOP the stage, record the failure verbatim in the run log, disposition per the matrix below, and only then re-run. Deviations from this SOP require Operator approval recorded in the run log.

Symptom	Likely cause	Disposition
Watchdog trip (exit 97/98)	memory pressure or a run-away solve	Re-establish the memory baseline (§7); reduce the mesh or defer; never retry immediately.
Code fixer exhausts attempts	a structural error the model cannot restructure	Inspect the saved fix attempts; if they are byte-identical the stall guard is not engaging — file a defect; otherwise re-run with a refined prompt or repair manually.
VLM CRITIQUE FAILED (both tries)	truncated or non-JSON output (context or token budget too small)	Check the error's raw-output snippet; verify Ollama health, the 8192-token context, and reference-image sizes; do NOT treat designer output from such a run as a valid iteration.
Critic rejects a visibly correct render	missing or wrong reference grounding for the family	Verify retrieval routing for the family (woven and Bouligand references rank first for their prompts); extend the image corpus additively.
Designer iterations churn without geometry change	critique not actionable, or cosmetic edits	If renders are pixel-identical across iterations, end the run (stall) and review the critique text being fed forward.
Audit: thin-feature failure	fibers below the process limit (as-designed demonstrators: 1.5–98% of material < 0.8 mm)	Stage 6 uniform scaling ($k \geq d_{\text{limit}}/d_{\text{certified}}$); re-certify.

Audit: (resin)	enclosed voids	closed pockets (micro-pockets at fiber weld lines)	For FDM: no action. For resin: add drain holes or reorient; micro-pockets ≤ 0.01 mL may be accepted with justification.
FEA match	lattice/linear mis-	mesh too coarse, plates not bonded (check <code>--overlap</code>), or a real regression	Verify the sidecar A_0/H_0 ; re-run the block validation; if the block fails, bisect the last solver change. Do not change a solver to fix a suspected defect without first reproducing it: a suspected reaction-extraction defect in the finite-strain solver was refuted by full-resolution cross-checks against the linear reference.

A Command quick reference

```
# memory snapshot sysctl -n vm.swapusage ; vm_stat | head -6
# fast tests python -m pytest -q -m "not slow"
# capped Ollama OLLAMA_MAX_LOADED_MODELS=1 OLLAMA_NUM_PARALLEL=1 OLLAMA_CONTEXT_LENGTH=8192 ollama serve
# full loop (watchdogged) <watchdog> 5120 2700 log -- <agent> "<prompt>"
# reproducible loop test <watchdog> ... -- <agent> --use-code saved_code.py "<prompt>"
# print audit (one part) "$ABCAD_CAD_PYTHON" -m abcad.printing.print_audit --stl X.stl [--voxel 0.15]
# FDM variants (+plated specimen) "$ABCAD_CAD_PYTHON" -m abcad.printing.fdm_variants --stl X.stl --scales 1.5,2.0 --plate 2.0 --overlap 0.8
# FEA linear reference "$ABCAD_SFEPY_PYTHON" -u abcad/fea/run_tension.py --msh X.msh
# FEA finite strain + self-check "$ABCAD_SFEPY_PYTHON" -u -m abcad.fea.run_tension_finite --msh X.msh | --block 10,10,20,5,5,9
```

The repository's `justfile` wraps these commands (`just --list`).

B The memory watchdog

`<watchdog> THRESH_MB MAX_SECONDS LOGFILE -- cmd [args...]` runs `cmd` in its own process group, samples swap 'used' and child-tree resident memory every 2s, appends each sample to `LOG-FILE`, and kills the whole group if swap exceeds the threshold (abort before the host stalls) or the time cap elapses (exit 97 / 98 respectively; the child's own exit code otherwise). The watchdog is a small host-side shell script and is intentionally not part of the package: thresholds are chosen per host and per step.

C Reference results anchoring this SOP's acceptance values

- Finite-strain reaction validation: $E_{\text{eff}} = 93.1$ MPa vs linear 91.8 MPa (+1.4%) on the 17,685-hex woven mesh; reaction identical to an independent assembly without essential boundary conditions; interior residual 7×10^{-7} N. Block: 3080.8 MPa (102.7%).

- Loop: structural repair in ≤ 2 code-fixer attempts, reproduced 3 $\times$; crash-free end to end; worst swap peak 1511 MB against a 5120 MB threshold over five instrumented runs. A complete run (repair $\rightarrow$ approval $\rightarrow$ deep-gate thin-feature failure $\rightarrow$ $\times 1.6$ auto-scale $\rightarrow$ PRINT) is archived in **results/agent_runs/**.
- Fleet audit: 10/10 demonstrator parts watertight and manifold; resin-printable 10/10 (2 with minor fixes); FDM-printable as designed 0/10 (thin features).
- Certified FDM variants of the 29 mm woven specimen: $\times 2.0$ clean pass (support-free, 58.2 mm, 31.1 mL); $\times 2.0$ plated specimen printable with top-plate supports or bridging; release candidate $\times 2.0$ remeshed: PRINT, lost@0.8 mm = 0.17%, zero enclosed voids.